

Learning Wiretap Codes with Neural Network Autoencoders for Keyless Physical-Layer Security

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Abstract:

Secure short-packet wireless links are increasingly expected to deliver two goals at the same time: high reliability for the intended receiver and strong confidentiality against passive interception. In many practical settings, especially for resource-constrained devices, conventional key-based security can be difficult to deploy because key distribution, management, and signaling overhead add complexity and latency. This challenge motivated our project, where we explore keyless physical-layer security through learned wiretap coding that can adapt its operating point as reliability and secrecy requirements change.

We design and evaluate a deep learning based wiretap communication system for degraded Gaussian wiretap channels. Alice maps k confidential bits together with r random “confusion” bits into a length- n real-valued codeword that is transmitted over an additive white Gaussian noise channel. Bob observes the main channel output and decodes the message, while Eve passively observes a noisier wiretap channel. Eve is assumed to know the channel model and the encoding strategy, so secrecy must be achieved through stochastic encoding and the channel advantage rather than obscurity or shared keys.

Our implementation uses TensorFlow and Keras with an autoencoder architecture that jointly learns an encoder (transmitter) and a decoder (Bob). A normalization layer enforces the transmit power constraint, and explicit Gaussian noise layers simulate Bob and Eve channels under configurable signal-to-noise ratios. Following the multi-objective formulation used in prior autoencoder wiretap code design, we optimize a weighted sum of two losses: (1) a mean-squared error reconstruction loss as a differentiable proxy for Bob’s block error behavior, and (2) an information leakage surrogate based on Gaussian mixture entropy bounds that yields an upper bound on leakage. We conduct systematic sweeps over the loss weights to trace the reliability secrecy Pareto tradeoff for fixed code parameters, and we export the learned codebook to compute leakage metrics consistently at evaluation time. We also include a polar wiretap coding baseline to contextualize performance.

Experimental results confirm the expected tradeoff: emphasizing secrecy reduces the leakage upper bound but increases Bob’s error rates. Across our sweep, leakage decreases from approximately 1.6 bits to about 0.85 bit, while BER increases from roughly 0.08 to about 0.23 and BLER rises from roughly 0.30 to about 0.60. These results demonstrate that the learned design can move smoothly between operating points, providing a practical mechanism to tune confidentiality versus reliability without changing blocklength or retraining entirely separate coding schemes. As interns, we contributed the end-to-end training pipeline, custom leakage loss implementation, weight-sweep experimentation, and performance evaluation using BER, BLER, and leakage upper bounds, establishing a reusable research codebase for future extensions to alternative leakage objectives, larger blocklengths, and more realistic channel models.